

L13 — Pointers and Pointer Arrays

Unit: 1 | Chapter: Pointers, Records & Advanced Arrays | BT Level: BT3 | CO: CO2

Learning Objectives

- Explain what a pointer is and how it relates to memory addresses
 - Describe pointer arithmetic and dereferencing
 - Implement pointer arrays and understand their use cases
 - Distinguish between shallow copy and deep copy using pointers
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1. What is a Pointer?

A **pointer** is a variable that stores the **memory address** of another variable rather than a direct value.

Memory Layout:

Address: 1000	Value: 42	← int x = 42
Address: 2000	Value: 1000	← int *ptr = &x (ptr holds address of x)

In C:

```
int x = 42;
int *ptr = &x;    // ptr stores the address of x

printf("%d", x);    // 42        - value of x
printf("%p", ptr);  // 0x1000    - address stored in ptr
printf("%d", *ptr); // 42        - dereference: value at the address
```

Python note: Python doesn't expose raw pointers, but uses references internally. The `id()` function returns the memory address of an object.

2. Pointer Fundamentals

2.1 Declaration and Initialization (C)

```
int x = 10;
int *p = &x;    // p is a pointer to int, stores address of x
int **pp = &p;  // pp is a pointer to a pointer to int

*p = 20;        // Modify x through pointer → x is now 20
**pp = 30;      // Modify x through double pointer → x is now 30
```

2.2 Pointer Arithmetic

Pointers can be incremented/decremented — they move by the size of the pointed-to type.

```
int arr[5] = {10, 20, 30, 40, 50};
int *p = arr;      // p points to arr[0]

p++;               // p now points to arr[1] (moved 4 bytes for int)
*(p + 2) = 99;     // Modifies arr[3] (arr[1] + 2 positions)

printf("%d", p[1]); // 30 (equivalent to *(p+1))
```

Arithmetic rules:

- $\text{ptr} + k \rightarrow$ moves $k \times \text{sizeof}(\text{type})$ bytes forward
 - $\text{ptr} - k \rightarrow$ moves $k \times \text{sizeof}(\text{type})$ bytes backward
 - $\text{ptr1} - \text{ptr2} \rightarrow$ number of elements between two pointers
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3. Arrays and Pointers

In C/C++, the name of an array is a **pointer to its first element**:

```
int arr[5] = {1, 2, 3, 4, 5};
// arr == &arr[0]
// arr[i] == *(arr + i)

for (int i = 0; i < 5; i++) {
    printf("%d ", *(arr + i)); // Same as arr[i]
}
```

This is why arrays can be passed to functions as pointers with no overhead:

```
void print_array(int *arr, int n) {
    for (int i = 0; i < n; i++)
        printf("%d ", arr[i]);
}
```

4. Pointer Arrays

A **pointer array** is an array where each element is a pointer.

4.1 Array of Pointers to Integers

```
int a = 10, b = 20, c = 30;
int *pArr[3];
pArr[0] = &a;
pArr[1] = &b;
pArr[2] = &c;

for (int i = 0; i < 3; i++)
    printf("%d ", *pArr[i]); // 10 20 30
```

4.2 Array of Strings (char pointers)

```
char *names[] = {"Alice", "Bob", "Charlie", "Diana"};

for (int i = 0; i < 4; i++)
    printf("%s\n", names[i]);
```

Memory layout:

```
names[0] → "Alice\0"
names[1] → "Bob\0"
names[2] → "Charlie\0"
names[3] → "Diana\0"
```

Each pointer takes 8 bytes (64-bit), but strings can be different lengths → **flexible storage**.

5. Pointer Arrays vs 2D Arrays

Feature	2D Array <code>char a[4][8]</code>	Pointer Array <code>char *a[4]</code>
Memory	Fixed: $4 \times 8 = 32$ bytes	Variable per string
Access	<code>a[i][j]</code>	<code>a[i][j]</code> (same syntax)
Row lengths	All equal	Can differ
Modifiable	In-place	Redirect pointer
Use for strings	Wastes space	Efficient

6. Sorting a Pointer Array

Sorting a pointer array rearranges pointers without moving data — efficient for large records:

```
# Python equivalent of pointer array sorting
students = ["Charlie", "Alice", "Diana", "Bob"]

# Sort the "pointers" (references) by the values they point to
students.sort() # Sorts references, not the actual string data in memo
# ["Alice", "Bob", "Charlie", "Diana"]
```

In C:

```
#include <stdlib.h>
#include <string.h>

char *names[] = {"Charlie", "Alice", "Diana", "Bob"};
int n = 4;

int cmp(const void *a, const void *b) {
    return strcmp(*(char **)a, *(char **)b);
}

qsort(names, n, sizeof(char *), cmp);
// names now: ["Alice", "Bob", "Charlie", "Diana"]
```

This is much faster than moving entire records — only pointers (8 bytes) are swapped.

7. Dynamic Memory Allocation via Pointers

```
#include <stdlib.h>

// Allocate array at runtime
int n = 10;
int *arr = (int *)malloc(n * sizeof(int));

if (arr == NULL) {
    // Handle allocation failure
    exit(1);
}

// Use arr like a normal array
for (int i = 0; i < n; i++)
    arr[i] = i * i;

free(arr); // Always free after use to prevent memory leaks
arr = NULL; // Good practice: avoid dangling pointer
```

8. NULL Pointer and Dangling Pointers

Issue	Description	Prevention
NULL pointer	Pointer not initialized, points to address 0	Always initialize pointers
Dangling pointer	Pointer to freed memory	Set to NULL after <code>free()</code>
Wild pointer	Uninitialized pointer with garbage address	Always initialize before use
Memory leak	Allocated but never freed	Call <code>free()</code> for every <code>malloc()</code>

9. Python Reference Model (Conceptual Pointers)

Python uses **references** (conceptually similar to pointers):

```
x = [1, 2, 3]
y = x          # y references the SAME list (shallow copy)
y.append(4)
print(x)       # [1, 2, 3, 4] - x is also modified

import copy
z = copy.deepcopy(x) # Deep copy - independent copy
z.append(5)
print(x)       # [1, 2, 3, 4] - x unchanged
```

Summary

- A **pointer** stores a memory address, allowing indirect access to data.
 - **Pointer arithmetic** moves by multiples of the element size.
 - Array name = pointer to first element in C.
 - **Pointer arrays** hold addresses of other data — useful for variable-length data and efficient sorting of large records.
 - In Python, all variables are references (conceptual pointers).
 - Common pitfalls: NULL pointers, dangling pointers, memory leaks.
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References

- Cormen et al., *Introduction to Algorithms*, Ch. 10.3 (MIT Press, 2022)
- Skiena, *The Algorithm Design Manual*, Ch. 3.1 (Springer, 2020)